

WASHINGTON, IA, USA

WMO: 725454

Lat: 41.276N

Lon: 91.673W

Elev: 754

StdP: 14.30

Time Zone: -6.00 (NAC)

Period: 95-19

WBAN: 04903

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.7	1.4	-10.7	3.2	-3.2	-6.1	4.1	3.8	30.5	17.3	27.1	21.8	9.3	290	0.538

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.6	92.8	77.8	90.4	77.0	88.0	75.3	81.6	88.6	79.4	86.8	77.6	85.2	9.7	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.4	156.9	86.3	77.1	145.2	84.4	75.0	135.1	82.6	45.8	89.2	43.6	86.8	41.4	85.4	90.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
26.0	23.0	19.9	-11.9	98.4	6.9	8.5	-16.8	104.6	-20.8	109.6	-24.7	114.4	-29.7	120.6		
			-10.0	83.6	8.7	2.8	-16.2	85.6	-21.3	87.2	-26.2	88.8	-32.5	90.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	51.4	24.6	27.7	39.7	52.1	63.2	72.4	75.2	73.3	65.2	53.6	40.0	28.7
	DBStd	20.41	12.45	12.14	11.74	9.74	8.48	6.20	6.03	5.52	7.96	9.90	10.55	11.37
	HDD50	2940	788	629	359	88	4	0	0	0	3	74	330	665
	HDD65	5985	1253	1046	786	400	138	11	3	6	91	376	749	1126
	CDD50	3465	0	3	40	149	415	673	782	724	460	185	31	3
	CDD65	1039	0	0	3	12	83	235	320	264	98	23	1	0
	CDH74	10142	0	0	23	183	838	2205	3271	2339	1044	232	7	0
	CDH80	3723	0	0	1	42	263	825	1349	821	362	60	0	0
Wind	WSAvg	9.2	10.8	10.7	11.2	11.6	10.1	8.2	6.4	5.8	6.8	8.9	10.4	10.1
Precipitation	PrecAvg	37.9	1.4	1.2	2.5	3.6	4.8	4.6	4.6	4.0	4.1	3.0	2.4	1.7
	PrecMax	60.6	4.2	4.1	5.9	9.0	11.5	11.2	12.8	11.8	12.9	8.0	7.2	6.3
	PrecMin	20.9	0.0	0.0	0.3	0.6	1.2	0.4	0.5	0.3	0.0	0.1	0.0	0.3
	PrecStd	8.5	1.0	0.9	1.5	2.1	2.8	2.5	2.7	2.4	2.8	2.0	1.7	1.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.0	63.6	77.4	85.9	91.1	93.5	98.6	93.5	92.9	86.4	73.4	63.0
		MCWB	48.0	52.8	61.2	67.9	72.3	76.8	78.6	78.9	75.0	71.5	59.4	57.5
	2%	DB	50.4	55.2	70.2	80.6	86.2	90.9	93.1	90.5	88.4	80.9	67.7	54.5
		MCWB	45.4	47.5	57.9	62.5	69.3	76.4	79.2	78.4	73.5	65.9	56.8	49.8
	5%	DB	44.9	49.6	63.4	74.6	82.0	88.3	90.6	88.0	84.1	75.0	62.6	48.3
		MCWB	40.6	45.0	54.2	59.9	66.7	74.3	78.4	76.9	71.6	64.2	53.9	44.6
	10%	DB	39.4	43.3	57.3	69.9	78.9	84.5	87.8	84.4	80.8	70.2	56.6	43.3
		MCWB	36.1	39.1	49.0	56.8	65.0	72.7	76.9	74.8	68.4	60.6	49.0	39.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.3	55.4	64.2	69.2	76.7	82.7	84.7	82.9	78.5	73.5	63.1	59.2
		MCDB	53.6	63.5	74.4	80.1	84.7	89.3	91.3	91.5	87.7	82.8	69.3	62.9
	2%	WB	45.0	49.3	59.9	64.8	73.2	79.1	82.4	80.2	76.0	69.7	58.7	51.0
		MCDB	49.9	55.6	67.2	76.2	82.6	86.6	89.3	87.9	85.2	75.4	65.3	53.2
	5%	WB	40.7	44.3	55.5	61.8	70.6	76.9	80.2	78.2	73.2	66.0	54.4	44.8
		MCDB	45.0	49.2	63.1	71.6	79.2	85.0	87.5	85.4	81.1	72.2	61.2	47.5
	10%	WB	36.9	39.4	50.2	58.8	67.8	74.8	78.2	76.3	70.8	62.4	49.9	39.7
		MCDB	39.7	43.6	57.1	67.8	75.2	82.5	85.6	83.1	78.0	68.9	55.8	43.2
Mean Daily Temperature Range	5% DB	MDBR	16.0	16.2	18.7	21.9	20.9	19.5	19.6	19.7	23.2	21.1	18.3	14.7
		MCDBR	21.4	24.4	26.9	30.1	25.1	22.6	21.4	22.0	26.8	26.7	25.8	20.4
		MCWBR	16.7	17.4	16.5	16.4	12.2	10.9	11.2	12.6	13.2	14.3	17.0	16.0
	5% WB	MCDBR	20.8	22.7	24.3	25.2	21.9	19.8	19.4	19.3	22.3	21.3	23.6	19.1
		MCWBR	16.5	17.0	16.0	16.1	12.9	11.9	12.6	12.4	12.5	13.6	17.4	16.3
Clear-Sky Solar Irradiance	taub	0.289	0.304	0.338	0.383	0.423	0.422	0.440	0.422	0.389	0.342	0.301	0.290	
	taud	2.427	2.390	2.385	2.269	2.223	2.279	2.211	2.274	2.322	2.454	2.501	2.443	
	Ebn at Noon	276	288	289	281	272	271	265	266	268	271	270	265	
	Edh at Noon	25	30	34	41	44	42	44	41	36	28	23	23	
All-Sky Solar Radiation	RadAvg	583	895	1191	1501	1762	1961	1963	1725	1468	975	655	484	
	RadStd	66	73	99	118	126	193	137	102	113	109	55	56	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (42 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page